

QUIZ #2 (55 PTS TOTAL)

**NOTE: Reference equations on the last page.
Good luck!**

1. (4 PTS – 0.5pt each) Health Physics – Matching:Column AColumn B

<u>G</u>	1. Roentgen (R)	A. Absorbed dose in energy (ergs or MeV) per gram of tissue .
<u>C</u>	2. RAD	B. Sensitive to $\beta/\gamma/x$ and utilized primarily for area radiation surveys.
<u>A</u>	3. REM	C. Absorbed dose in energy (ergs or MeV) per gram of matter.
<u>D</u>	4. Curie (Ci)	D. Amount of radioactivity in 1 gram of Radium.
	Becquerel	E. 1 decay per second (DPS).
<u>E</u>	5. (Bq)	
	Geiger	F. Sensitive to $\alpha/\beta/\gamma/x$ and utilized primarily for contamination surveys.
<u>F</u>	6. Counter	
<u>B</u>	7. Ion Chamber	G. Exposure in coulombs/cc of air.
<u>H</u>	8. REM Ball	H. Sensitive to fast neutrons; reads out in mREM/hr.

2. (6 PTS) You have a 1 Ci source of Cesium-137. The primary gamma decay energy is 661.7 keV, with a likelihood of 94.7%. What is the exposure rate (R/hr) at a distance of 1 foot? Using the previously calculated dose rate and the **inverse square law**, what would the exposure rate be at 5 feet from the same source? Show your work.

(2PT Setup) $DR_1 = 6 \cdot C \cdot E \cdot n / d^2 = 6 \cdot 1 \text{ Ci} \cdot 0.6617 \text{ MeV} \cdot 0.947 / (1\text{ft})^2$

(1.5PT Answer) $DR_1 = 3.7598 \text{ R/hr}$

(1.5PT Setup) $DR_1 \cdot R_1^2 = DR_2 \cdot R_2^2$; $DR_2 = DR_1 \cdot R_1^2 / R_2^2$

(1PT Answer) $DR_2 = 3.7598 \text{ R/hr} \cdot (1\text{ft}^2) / (5\text{ft}^2) = 0.1504 \text{ R/hr}$

3. During a reactor startup, the initial NFM count-rate $CR_0 = 15$ CPS at time zero ($t=0$) and the initial value for $\rho_{\text{net}} = -8100$ pcm with the control rods seated (reactor fully shutdown).

- a) (8 PTS) Fill in the values for K_{eff} and reactivity ρ in the following table as rods are withdrawn and the NFM count-rate is approximately doubled 5X – **SHOW ALL WORK!**
Calculate k_{eff} and ρ values to 5 decimal places (e.g. 0.00001):

Time	NFM (CPS)	K_{eff}	Reactivity ρ_{net} ($\Delta k/k$)
0	$CR_0 = 15$	0.92507	-0.08100 (-8100 pcm)
1	$CR_1 = 35$	0.96789	-0.03318 (-3318 pcm)
2	$CR_2 = 80$	0.98595	-0.01425 (-1425 pcm)
3	$CR_3 = 180$	0.99376	-0.00628 (-628 pcm)
4	$CR_4 = 600$	0.99813	-0.00188 (-188 pcm)
5	$CR_5 = 4000$	0.99972	-0.00028 (-28 pcm)

$$0: K_0 = \frac{1}{1 - (-0.08100)} = 0.92507$$

$$1: K_1 = 1 - \frac{CR_0}{CR_1} (1 - K_0) = 1 - \frac{15}{35} (1 - 0.92507) = 0.96789; \rho = \frac{0.96789 - 1}{0.96789} = -0.03318$$

$$2: K_2 = 1 - \frac{CR_1}{CR_2} (1 - K_1) = 1 - \frac{35}{80} (1 - 0.96789) = 0.98595; \rho = \frac{0.98595 - 1}{0.98595} = -0.01425$$

$$3: K_3 = 1 - \frac{CR_2}{CR_3} (1 - K_2) = 1 - \frac{80}{180} (1 - 0.98595) = 0.99376; \rho = \frac{0.99376 - 1}{0.99376} = -0.00628$$

$$4: K_4 = 1 - \frac{CR_3}{CR_4} (1 - K_3) = 1 - \frac{180}{600} (1 - 0.99376) = 0.99813; \rho = \frac{0.99813 - 1}{0.99813} = -0.00188$$

$$5: K_5 = 1 - \frac{CR_4}{CR_5} (1 - K_4) = 1 - \frac{600}{4000} (1 - 0.99813) = 0.99972; \rho = \frac{0.99972 - 1}{0.99972} = -0.00028$$

Split points up however you'd like for this question, as long as it's consistent. Small rounding errors should not deduct points.

- b) (5 PTS) At timestep #5 (after the 5th double), is the reactor super-critical or sub-critical?
By how much net reactivity ρ_{net} (pcm)?

Subcritical by -28 pcm

Go based on the answer they gave for time 5 in part a, even if that answer was incorrect.

4. (9 PTS, 1 Each) Answer TRUE or FALSE to the following:

- T a. The prompt neutron generation time (ℓ_p) is less than the delayed neutron generation time (ℓ_d).
- F b. The prompt neutron fraction ($1-\beta$) is less than the delayed neutron fraction (β).
- F c. Equal changes in K_{eff} result in equal changes in the equilibrium neutron population N_{eq} for a subcritical reactor.
- F d. As K_{eff} approaches 1.0, more generations are required to reach equilibrium; therefore, less time is required to reach equilibrium.
- F e. If K_{eff} is increased halfway to 1.0 from its initial value, the neutron count rate will increase by half.
- F f. Reactor period τ is the time it takes for reactor power to change by a factor of 3.14 (or 'e').
- F g. RF, Microwave, and Infrared are examples of ionizing radiation.
- F h. In Stochastic health effects of radiation, the severity increases with dose.
- T i. Minimizing the time spent near a radiation source would be consistent with As Low As Reasonably Achievable (ALARA) principles.

5. (10 PTS) If $\ell_p = 3 \times 10^{-5}$ sec, $\ell_d = 12.7$ sec, $\beta = 0.0073$, and 100 pcm of reactivity is added to the core from an initially critical state (i.e. $\rho_{net} = +100$ pcm), calculate the final power in Watts (P_f) at $t = 60$ sec if $P_i = 250$ Watts.

(4 PTS) First: $\bar{l} = l_d(\beta) + l_p(1 - \beta)$; $\bar{l} = (12.7 \times 0.0073) + (3E - 5)(1 - 0.0073) = 0.09274$ sec

(2 PTS) Second: $K_{eff} = 1/(1-\rho) = 1/(1-0.0010) = 1.0010$

(4 PTS) Third: $P_f = P_o e^{\left(\frac{K_{eff}-1}{\bar{l}}\right)t}$; so $P_f = 250 e^{\left(\frac{1.0010-1}{0.09274}\right)60} = 477$ W

See lab questions on the next page

HP Lab Questions:

4. (2 PTS) What are the limits for a Radiation Area and a High Radiation Area in mrem/hr?

Radiation Area – Greater than or equal to 5 mrem/hr (at a distance of 30cm)

High Radiation Area - Greater than or equal to 100mrem/hr (at a distance of 30cm)

5. (2 PTS) What are two important pre-operational checks to verify before utilizing a hand-held radiation meter?

Perform a battery check and assure that the level is acceptable

Verify that the meter is within the calibration date

1/M Lab Questions:

(4 PTS) Answer TRUE or FALSE to the following:

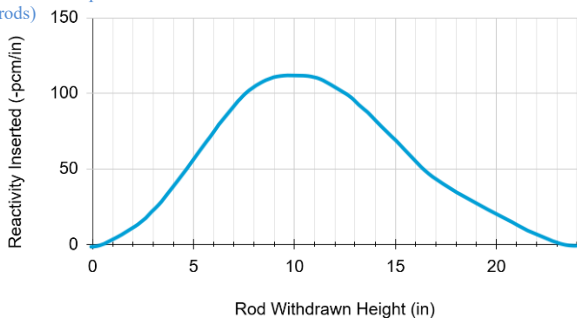
- | | |
|----------|---|
| <u>F</u> | a. We increased the number of fission neutron (N_f) in the reactor by lowering control rods into the core. |
| <u>T</u> | b. For each rod pull, we pulled <u>halfway</u> from where the pull started to the projected critical rod height OR until NFM counts doubled, whichever happened first. |
| <u>T</u> | c. As $CR_i \rightarrow \infty$, $M \rightarrow \infty$, $1/M \rightarrow 0$, and $K_{eff} \rightarrow 1$. |
| <u>T</u> | d. As we got closer to criticality, it took longer for the neutron population to reach subcritical equilibrium. |

Control Rod Calibration Lab Questions

1. (4 PTS) Draw a sketch of what the differential and integral worth may look like for a Control Rod at the PULSTAR reactor.:

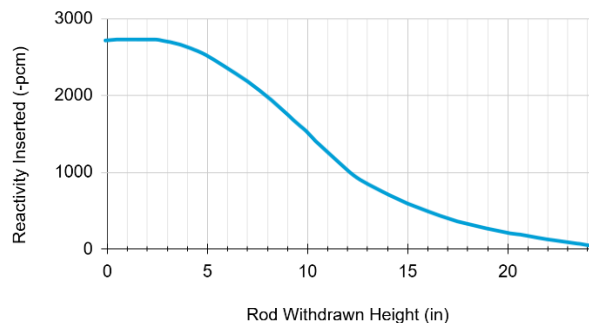
0.5 bonus (to a max of 100%) if they seem to intentionally place the peak of the differential under 12 inches (axial flux depression from other rods)

Differential Rod Worth



0 at both 0" and 24" withdrawn. The peak should be around 12". Should be a rough sinusoidal

Integral Rod Worth



Intercept at between -1000 and -3000 pcm. *near* Zero slope at 0" and 24" height Inflection point should be around 12"

2. (1 PTS) Which control rod did we calibrate during the lab?

- ☐ Safety #1
☐ Safety #2
☐ Regulating Rod
☐ Shim Rod

Equations:

$$\rho = (K_{\text{eff}} - 1) / K_{\text{eff}} \quad \text{where } \rho \text{ is in units of } \Delta k/k.$$

$$K_{\text{eff}} = 1/(1-\rho) \quad \text{where } \rho \text{ is in units of } \Delta k/k.$$

$$\beta_{\text{eff}} = 730 \text{ pcm} = 0.0073 \Delta k/k \text{ (assumed value for PULSTAR)}$$

$$CR_2/CR_1 = (1-K_{\text{eff}1})/(1-K_{\text{eff}2})$$

$$\bar{l} = l_d(\beta) + l_p(1 - \beta)$$

$$P_f = P_o e^{\left(\frac{K_{\text{eff}} - 1}{\bar{l}}\right)t}, \text{ for small reactivity insertions}$$

$$1 \text{ Ci} = 3.7 \times 10^{10} \text{ dis/sec}$$

$$1 \text{ kg} = 2.21 \text{ lbm}$$

$$1 \text{ MW} = 3.41 \times 10^6 \text{ BTU/hr}$$

$$N_A = 6.022 \times 10^{23} \text{ atoms/mol}$$

$$DR = \frac{6 * C * E * n}{d^2}, \text{ where:}$$

DR: Dose rate in R/hr

C: Activity in Curies

E: Energy in MeV

n: Probability of occurrence per decay

d: Distance in ft